WORKED SOLUTION

STEP II 1988, Question 7

Pure • Integration by Substitution

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/88-s2-q7

The key idea

Every equality here is a substitution, and the hard part is finding the right one. The first step needs a substitution that sends $[1,2]$ to $[0,1]$ and turns $2-2x+x^2$ into something with $1+t^2$ in it: a Möbius map $x = 2/(1+t)$ does exactly that. Once $1+t^2$ appears, $t=\tan\theta$ is the obvious move. The θ-integrand is symmetric about $\theta = \frac{1}{8}\pi$, and undoing $t=\tan\theta$ on the half-range brings in $\tan\frac{1}{8}\pi = \sqrt{2}-1$. The final deduction uses the same substitutions in reverse.

Solution

Throughout, k is any real constant. All the bases we raise to powers are positive on the ranges involved, so non-integer powers cause no trouble.

From $[1,2]$ to $[0,1]$

Put $x = \dfrac{2}{1+t}$, so that $t=1$ when $x=1$, $t=0$ when $x=2$, and $\mathrm{d}x = -\dfrac{2}{(1+t)^2}\,\mathrm{d}t$. Then

$$2-2x+x^2 = \frac{2(1+t)^2-4(1+t)+4}{(1+t)^2} = \frac{2(1+t^2)}{(1+t)^2}, \qquad x^{k+1} = \frac{2^{k+1}}{(1+t)^{k+1}}.$$

So the integrand becomes

$$\frac{2^k(1+t^2)^k}{(1+t)^{2k}} \cdot \frac{(1+t)^{k+1}}{2^{k+1}} \cdot \frac{2}{(1+t)^2} = \frac{(1+t^2)^k}{(1+t)^{k+1}},$$

and the minus sign in $\mathrm{d}x$ swaps the limits from $\int_1^0$ to $\int_0^1$:

$$I = \int_0^1 \frac{(1+t^2)^k}{(1+t)^{k+1}}\,\mathrm{d}t.$$

The trigonometric form

Now put $t = \tan\theta$, with θ running from 0 to $\frac{1}{4}\pi$. Then $1 + t^2 = \sec^2\theta$, $\mathrm{d}t = \sec^2\theta\,\mathrm{d}\theta$, and

$$1 + t = \frac{\cos\theta + \sin\theta}{\cos\theta} = \frac{\sqrt{2}\cos\left(\frac{1}{4}\pi - \theta\right)}{\cos\theta},$$

using $\cos\theta + \sin\theta = \sqrt{2}\left(\cos\frac{1}{4}\pi\cos\theta + \sin\frac{1}{4}\pi\sin\theta\right)$. The integrand becomes

$$\sec^{2k}\theta \cdot \frac{\cos^{k+1}\theta}{\left[\sqrt{2}\cos\left(\frac{1}{4}\pi - \theta\right)\right]^{k+1}} \cdot \sec^2\theta = \frac{\cos^{-(k+1)}\theta}{\left[\sqrt{2}\cos\left(\frac{1}{4}\pi - \theta\right)\right]^{k+1}} = \frac{1}{\left[\sqrt{2}\cos\theta\cos\left(\frac{1}{4}\pi - \theta\right)\right]^{k+1}},$$

since the powers of $\cos\theta$ total $-2k + (k+1) - 2 = -(k+1)$. Both cosines are positive on $[0, \frac{1}{4}\pi]$, so this is valid for every real k. That gives the second form of I.

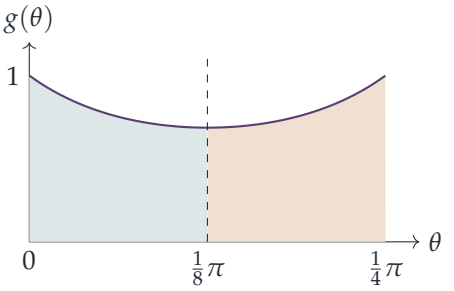
Ask yourself Why should the integral over $[0, \frac{1}{4}\pi]$ be twice the integral over $[0, \frac{1}{8}\pi]$?

Write $g(\theta)$ for the θ-integrand. Replacing θ by $\frac{1}{4}\pi - \theta$ just swaps the two factors $\cos\theta$ and $\cos(\frac{1}{4}\pi - \theta)$, so $g(\frac{1}{4}\pi - \theta) = g(\theta)$: the graph of g is symmetric about $\theta = \frac{1}{8}\pi$. (You can see this directly too: $\sqrt{2}\cos\theta\cos(\frac{1}{4}\pi - \theta) = \frac{1}{2}(1 + \sqrt{2}\cos(2\theta - \frac{1}{4}\pi))$, which depends only on the distance of θ from $\frac{1}{8}\pi$.) Formally, substituting $\phi = \frac{1}{4}\pi - \theta$,

$$\int_{\pi/8}^{\pi/4} g(\theta)\,\mathrm{d}\theta = \int_0^{\pi/8} g(\tfrac{1}{4}\pi - \phi)\,\mathrm{d}\phi = \int_0^{\pi/8} g(\phi)\,\mathrm{d}\phi,$$

so the two halves of $[0, \frac{1}{4}\pi]$ contribute equally and

$$I = 2\int_0^{\pi/8} \frac{\mathrm{d}\theta}{\left[\sqrt{2}\cos\theta\cos\left(\frac{1}{4}\pi - \theta\right)\right]^{k+1}}.$$



The figure shows g for $k = 1$: the two shaded halves are mirror images.

Hence the integral up to $\sqrt{2}-1$

Now run the substitution $t=\tan\theta$ backwards on $[0,\frac{1}{8}\pi]$. Every step of the computation above is unchanged; only the upper limit moves, to $t=\tan\frac{1}{8}\pi$. From $\tan\frac{1}{4}\pi = \dfrac{2\tau}{1-\tau^2}=1$ with $\tau=\tan\frac{1}{8}\pi$ we get $\tau^2+2\tau-1=0$, so $\tau=-1\pm\sqrt{2}$, and the positive root is $\tan\frac{1}{8}\pi=\sqrt{2}-1$. Hence

$$I=2\int_0^{\sqrt{2}-1}\frac{(1+t^2)^k}{(1+t)^{k+1}}\,\mathrm{d}t,$$

as required. Comparing this with $I=\int_0^1$ of the same integrand, we have found that the integral over $[0,\sqrt{2}-1]$ and the integral over $[\sqrt{2}-1,1]$ are *both* equal to $\frac{1}{2}I$. This is the fact we use next.

The deduction

Left-hand side. Write $\dfrac{2-2x^2+x^4}{x^2}=\dfrac{2-2u+u^2}{u}$ with $u=x^2$. Then $\mathrm{d}u=2x\,\mathrm{d}x$, so $\dfrac{\mathrm{d}x}{x}=\dfrac{\mathrm{d}u}{2u}$, and u runs from 1 to 2:

$$\int_1^{\sqrt{2}}\left(\frac{2-2x^2+x^4}{x^2}\right)^k\frac{1}{x}\,\mathrm{d}x=\frac{1}{2}\int_1^2\frac{(2-2u+u^2)^k}{u^k}\cdot\frac{1}{u}\,\mathrm{d}u=\frac{1}{2}I.$$

Right-hand side. The integrand is $\dfrac{(2-2x+x^2)^k}{x^{k+1}}$, the same as in the definition of I, but over $[1,\sqrt{2}]$. Use the first substitution $x=2/(1+t)$ again. Now $x=\sqrt{2}$ corresponds to $1+t=\sqrt{2}$, that is $t=\sqrt{2}-1$, so

$$\begin{aligned}\int_1^{\sqrt{2}}\left(\frac{2-2x+x^2}{x}\right)^k\frac{1}{x}\,\mathrm{d}x&=\int_{\sqrt{2}-1}^1\frac{(1+t^2)^k}{(1+t)^{k+1}}\,\mathrm{d}t\\&=I-\int_0^{\sqrt{2}-1}\frac{(1+t^2)^k}{(1+t)^{k+1}}\,\mathrm{d}t=I-\tfrac{1}{2}I=\tfrac{1}{2}I.\end{aligned}$$

Both sides equal $\frac{1}{2}I$, so they are equal, which is the required result.

Checks

- $k=0$: $I=\int_1^2\frac{1}{x}\,\mathrm{d}x=\ln 2$, and both sides of the final identity are $\int_1^{\sqrt{2}}\frac{1}{x}\,\mathrm{d}x=\frac{1}{2}\ln 2$, as they should be.
- $k=1$: $I=\int_1^2\left(\frac{2}{x^2}-\frac{2}{x}+1\right)\mathrm{d}x=2-2\ln 2\approx 0.61371$. Simpson's rule on all five forms of I, and on both sides of the final identity, agrees to ten decimal places (each side is ≈ 0.30685). The same agreement holds for non-integer and negative k, such as $k=0.5$, $k=-0.7$ and $k=4.3$.

Common slips

- Forgetting the factor $\sqrt{2}$ when writing $\cos\theta + \sin\theta$ as a single cosine, which leaves a stray $2^{(k+1)/2}$.
- Trying to evaluate I. For general k there is no closed form, and none is needed: every step is a change of variable or a symmetry.
- In the deduction, missing that the right-hand side is only *part* of I and needs the "hence" result to finish.

Beyond STEP

The map $x \mapsto 2/(1+t)$ is a Möbius (fractional linear) transformation. These maps send circles and lines to circles and lines, and they are the natural changes of variable for integrands built from quadratics. Spotting a hidden symmetry of an integrand, as with $\theta \mapsto \frac{1}{4}\pi - \theta$ here, is a standard trick for evaluating definite integrals that have no elementary antiderivative. For more on Möbius transformations, read Needham, *Visual Complex Analysis*. For many more integrals of this style, see Siklos, *Advanced Problems in Mathematics*.

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